

Questions

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1. The line ℓ_1 is the intersection of the planes

$$x + y + z = 1$$

and

$$2x - y + z = 4$$

The plane Π has equation

$$x + y + 2z = 3$$

The line ℓ_2 is the reflection of the line ℓ_1 in the plane Π .

Find a vector equation of the line ℓ_2 .

[7]

Solution

Write the reflecting plane in the form

$$\Pi : \mathbf{r} \cdot \mathbf{n} = d$$

so here

$$\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad d = 3$$

To find the reflection of the line ℓ_1 , we use the reflection formulae for a point and a direction vector in a plane.

First find a direction vector for ℓ_1 . Since ℓ_1 is the intersection of the planes

$$x + y + z = 1 \quad \text{and} \quad 2x - y + z = 4$$

their normal vectors are

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

So a direction vector of ℓ_1 is

$$\begin{aligned} \mathbf{d}_1 &= \mathbf{n}_1 \times \mathbf{n}_2 \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} \\ &= 2\mathbf{i} + \mathbf{j} - 3\mathbf{k} \\ &= \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} \end{aligned}$$

Now find a convenient point on ℓ_1 . The best choice is the point where ℓ_1 meets Π , since a point on the mirror plane is unchanged by reflection.

So solve

$$\begin{aligned} x + y + z &= 1 \\ 2x - y + z &= 4 \\ x + y + 2z &= 3 \end{aligned}$$

Subtracting the first equation from the third gives

$$z = 2$$

Substituting $z = 2$ into the first two equations:

$$\begin{aligned} x + y + 2 &= 1 \\ 2x - y + 2 &= 4 \end{aligned}$$

so

$$\begin{aligned}x + y &= -1 \\ 2x - y &= 2\end{aligned}$$

Adding these equations:

$$\begin{aligned}3x &= 1 \\ x &= \frac{1}{3}\end{aligned}$$

Then

$$y = -1 - \frac{1}{3} = -\frac{4}{3}$$

Hence the point is

$$\mathbf{q} = \begin{pmatrix} \frac{1}{3} \\ -\frac{4}{3} \\ 2 \end{pmatrix}$$

Using the point reflection formula

$$\mathbf{q}' = \mathbf{q} - 2 \frac{\mathbf{q} \cdot \mathbf{n} - d}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

we note that $\mathbf{q}$ lies on Π , so $\mathbf{q} \cdot \mathbf{n} = d$. Therefore

$$\mathbf{q}' = \mathbf{q}$$

So the reflected line ℓ_2 passes through the same point

$$\begin{pmatrix} \frac{1}{3} \\ -\frac{4}{3} \\ 2 \end{pmatrix}$$

Now reflect the direction vector using

$$\mathbf{d}_2 = \mathbf{d}_1 - 2 \frac{\mathbf{d}_1 \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

Compute the dot products:

$$\begin{aligned}\mathbf{d}_1 \cdot \mathbf{n} &= \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 2 + 1 - 6 = -3 \\ \mathbf{n} \cdot \mathbf{n} &= \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = 1 + 1 + 4 = 6\end{aligned}$$

Hence

$$\begin{aligned}\mathbf{d}_2 &= \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} - 2 \left(\frac{-3}{6} \right) \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}\end{aligned}$$

Therefore a vector equation of the reflected line ℓ_2 is

$$\mathbf{r} = \begin{pmatrix} \frac{1}{3} \\ -\frac{4}{3} \\ 2 \end{pmatrix} + t \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}, \quad t \in \mathbb{R}$$

This is a valid vector equation of the line ℓ_2 .

2. The plane Π has vector equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + s \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

The line ℓ_1 passes through the point $P(4, 1, 2)$ and is perpendicular to Π .

The line ℓ_1 meets Π at the point Q .

(a) Find the coordinates of Q . [4]

Given that the point $R(3, 1, 0)$ lies on Π ,

(b) find a Cartesian equation of the plane containing P , Q and R . [4]

The line ℓ_2 passes through P and R .

The line ℓ_3 is the reflection of ℓ_2 in Π .

(c) Find a vector equation of ℓ_3 . [4]

Solution

(a) The plane Π has direction vectors

$$\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

So a normal vector to Π is

$$\begin{aligned} \mathbf{n} &= \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} (-1)(0) - 1(1) \\ 1(1) - 2(0) \\ 2(1) - (-1)(1) \end{pmatrix} \\ &= \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \end{aligned}$$

Using the point $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ on Π ,

$$\begin{aligned} d &= \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \\ &= -1 + 2 - 3 \\ &= -2 \end{aligned}$$

Hence Π is

$$\mathbf{r} \cdot \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} = -2$$

so in Cartesian form,

$$-x + y + 3z + 2 = 0$$

Since ℓ_1 is perpendicular to Π , it has direction vector $\mathbf{n}$. Therefore

$$\ell_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$$

Using the line-plane formula

$$\lambda_* = \frac{d - \mathbf{a} \cdot \mathbf{n}}{\mathbf{b} \cdot \mathbf{n}}$$

$$\text{with } \mathbf{a} = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}, d = -2,$$

$$\begin{aligned} \lambda &= \frac{-2 - \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}}{\begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}} \\ &= \frac{-2 - (-4 + 1 + 6)}{1 + 1 + 9} \\ &= \frac{-2 - 3}{11} \\ &= -\frac{5}{11} \end{aligned}$$

So

$$\begin{aligned} Q &= \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} - \frac{5}{11} \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 4 + \frac{5}{11} \\ 1 - \frac{5}{11} \\ 2 - \frac{15}{11} \end{pmatrix} \\ &= \begin{pmatrix} \frac{49}{11} \\ \frac{6}{11} \\ \frac{7}{11} \end{pmatrix} \end{aligned}$$

Hence

$$Q \left(\frac{49}{11}, \frac{6}{11}, \frac{7}{11} \right)$$

(b) A plane containing P, Q, R contains the vectors $\overrightarrow{PQ}$ and $\overrightarrow{QR}$.

First,

$$\begin{aligned} \overrightarrow{PQ} &= \begin{pmatrix} \frac{49}{11} \\ \frac{6}{11} \\ \frac{7}{11} \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{5}{11} \\ -\frac{5}{11} \\ -\frac{15}{11} \end{pmatrix} = \frac{5}{11} \begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix} \end{aligned}$$

Also,

$$\begin{aligned}\overrightarrow{QR} &= \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} \frac{49}{11} \\ \frac{6}{11} \\ \frac{7}{11} \end{pmatrix} \\ &= \begin{pmatrix} -\frac{16}{11} \\ \frac{5}{11} \\ -\frac{7}{11} \end{pmatrix} = \frac{1}{11} \begin{pmatrix} -16 \\ 5 \\ -7 \end{pmatrix}\end{aligned}$$

So a normal vector to the plane PQR is

$$\begin{aligned}\mathbf{n}_1 &= \begin{pmatrix} 1 \\ -1 \\ -3 \end{pmatrix} \times \begin{pmatrix} -16 \\ 5 \\ -7 \end{pmatrix} \\ &= \begin{pmatrix} (-1)(-7) - (-3)(5) \\ (-3)(-16) - 1(-7) \\ 1(5) - (-1)(-16) \end{pmatrix} \\ &= \begin{pmatrix} 22 \\ 55 \\ -11 \end{pmatrix} \\ &= 11 \begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix}\end{aligned}$$

Hence we may take the normal vector as

$$\begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix}$$

Using the point $P(4, 1, 2)$, the plane equation is

$$\begin{pmatrix} 2 \\ 5 \\ -1 \end{pmatrix} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \right) = 0$$

So

$$\begin{aligned}2(x - 4) + 5(y - 1) - (z - 2) &= 0 \\ 2x - 8 + 5y - 5 - z + 2 &= 0 \\ 2x + 5y - z - 11 &= 0\end{aligned}$$

Therefore a Cartesian equation of the plane containing P, Q, R is

$$2x + 5y - z - 11 = 0$$

(c) The line ℓ_2 passes through $R(3, 1, 0)$ and $P(4, 1, 2)$, so

$$\ell_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

Write Π in the form $\mathbf{r} \cdot \mathbf{n} = d$, where

$$\mathbf{n} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}, \quad d = -2$$

Since R lies on Π , it is unchanged by reflection. Indeed,

$$\begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} = -3 + 1 + 0 = -2 = d$$

So to find ℓ_3 , we reflect the direction vector

$$\mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

Using the reflection formula for a direction vector,

$$\mathbf{b}' = \mathbf{b} - 2 \frac{\mathbf{b} \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

Now

$$\begin{aligned} \mathbf{b} \cdot \mathbf{n} &= \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \\ &= -1 + 0 + 6 \\ &= 5 \end{aligned}$$

and

$$\begin{aligned} \mathbf{n} \cdot \mathbf{n} &= \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \\ &= 1 + 1 + 9 \\ &= 11 \end{aligned}$$

Therefore

$$\begin{aligned} \mathbf{b}' &= \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - 2 \cdot \frac{5}{11} \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} -\frac{10}{11} \\ \frac{10}{11} \\ \frac{30}{11} \end{pmatrix} \\ &= \begin{pmatrix} \frac{21}{11} \\ -\frac{10}{11} \\ -\frac{8}{11} \end{pmatrix} \end{aligned}$$

This is parallel to $\begin{pmatrix} 21 \\ -10 \\ -8 \end{pmatrix}$. Hence

$$\ell_3 : \mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 21 \\ -10 \\ -8 \end{pmatrix}$$

So a vector equation of ℓ_3 is

$$\mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 21 \\ -10 \\ -8 \end{pmatrix}$$

3. A plane passes through the points $C(1, -1, 2)$, $D(3, 0, 1)$ and $E(2, 2, 2)$.

The point A has coordinates $(4, -2, 7)$ and the point B is the reflection of A in the plane.

Find the coordinates of the point B .

[7]

Solution

Let the plane be Π .

Two direction vectors in the plane are

$$\overrightarrow{CD} = (3 - 1, 0 - (-1), 1 - 2) = (2, 1, -1)$$

and

$$\overrightarrow{CE} = (2 - 1, 2 - (-1), 2 - 2) = (1, 3, 0)$$

So a normal vector to the plane is

$$\begin{aligned} \mathbf{n} &= \overrightarrow{CD} \times \overrightarrow{CE} \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -1 \\ 1 & 3 & 0 \end{vmatrix} \\ &= \mathbf{i}(1 \cdot 0 - (-1) \cdot 3) - \mathbf{j}(2 \cdot 0 - (-1) \cdot 1) + \mathbf{k}(2 \cdot 3 - 1 \cdot 1) \\ &= 3\mathbf{i} - \mathbf{j} + 5\mathbf{k} \\ &= (3, -1, 5) \end{aligned}$$

Using the form $\mathbf{r} \cdot \mathbf{n} = d$, we find d from point $C(1, -1, 2)$:

$$\begin{aligned} d &= (1, -1, 2) \cdot (3, -1, 5) \\ &= 1 \cdot 3 + (-1) \cdot (-1) + 2 \cdot 5 \\ &= 3 + 1 + 10 \\ &= 14 \end{aligned}$$

Hence the plane is

$$\mathbf{r} \cdot (3, -1, 5) = 14$$

so in Cartesian form,

$$3x - y + 5z - 14 = 0$$

Now use the reflection formula for a point in a plane:

$$\mathbf{p}' = \mathbf{p} - 2 \frac{\mathbf{p} \cdot \mathbf{n} - d}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

Here

$$\mathbf{p} = (4, -2, 7), \quad \mathbf{n} = (3, -1, 5), \quad d = 14$$

First calculate $\mathbf{p} \cdot \mathbf{n} - d$:

$$\begin{aligned} \mathbf{p} \cdot \mathbf{n} - d &= (4, -2, 7) \cdot (3, -1, 5) - 14 \\ &= 4 \cdot 3 + (-2) \cdot (-1) + 7 \cdot 5 - 14 \\ &= 12 + 2 + 35 - 14 \\ &= 35 \end{aligned}$$

Also,

$$\mathbf{n} \cdot \mathbf{n} = 3^2 + (-1)^2 + 5^2 = 9 + 1 + 25 = 35$$

Substituting into the reflection formula:

$$\begin{aligned} \mathbf{p}' &= (4, -2, 7) - 2 \left(\frac{35}{35} \right) (3, -1, 5) \\ &= (4, -2, 7) - 2(3, -1, 5) \\ &= (4, -2, 7) - (6, -2, 10) \\ &= (-2, 0, -3) \end{aligned}$$

Therefore the coordinates of B are

$$B = (-2, 0, -3)$$

As a quick check, the closest point on the plane to A is

$$\mathbf{p} - \frac{\mathbf{p} \cdot \mathbf{n} - d}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n} = (4, -2, 7) - \frac{35}{35}(3, -1, 5) = (1, -1, 2) = C$$

so C is the midpoint of AB , which confirms the result.

Final answer: $B = (-2, 0, -3)$.

4. A line ℓ has equation

$$\mathbf{r} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

Another line m has equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

A plane Π contains m and is parallel to ℓ .

- (i) Find an equation of Π , giving your answer in the form $\mathbf{r} \cdot \mathbf{n} = p$. [4]
- (ii) Find the distance between ℓ and Π . [4]
- (iii) Find an equation of the line which is the reflection of ℓ in Π , giving your answer in the form $\mathbf{r} = \mathbf{a} + t\mathbf{b}$. [4]

Solution

- (i) Let the direction vector of line ℓ be

$$\mathbf{b} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

and the direction vector of line m be

$$\mathbf{d} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

Since Π contains m and is parallel to ℓ , both $\mathbf{d}$ and $\mathbf{b}$ lie in the plane, so a normal vector is

$$\begin{aligned} \mathbf{n} &= \mathbf{d} \times \mathbf{b} \\ &= \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot 1 - 0 \cdot (-1) \\ -(2 \cdot 1 - 0 \cdot 2) \\ 2 \cdot (-1) - 1 \cdot 2 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} \end{aligned}$$

A point on m is $\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$, so for the plane equation $\mathbf{r} \cdot \mathbf{n} = p$,

$$\begin{aligned} p &= \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \\ &= 1 - 2 - 8 \\ &= -9 \end{aligned}$$

Hence the plane is

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} = -9$$

So an equation of Π is

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} = -9$$

(ii) From the line-plane formula, if $\mathbf{b} \cdot \mathbf{n} = 0$, then the line is parallel to the plane.

Here

$$\begin{aligned} \mathbf{b} \cdot \mathbf{n} &= \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} \\ &= 2 + 2 - 4 \\ &= 0 \end{aligned}$$

So $\ell \parallel \Pi$.

Take the point $P = (4, 2, -3)$ on ℓ . Using the point-to-plane distance formula

$$\text{dist} = \frac{|\mathbf{p} \cdot \mathbf{n} - d|}{|\mathbf{n}|}$$

with $\mathbf{p} = \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix}$, $\mathbf{n} = \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix}$ and $d = -9$,

$$\begin{aligned} \mathbf{p} \cdot \mathbf{n} &= \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} \\ &= 4 - 4 + 12 \\ &= 12 \end{aligned}$$

Also,

$$\begin{aligned} |\mathbf{n}| &= \sqrt{1^2 + (-2)^2 + (-4)^2} \\ &= \sqrt{1 + 4 + 16} \\ &= \sqrt{21} \end{aligned}$$

Therefore

$$\begin{aligned} \text{dist} &= \frac{|12 - (-9)|}{\sqrt{21}} \\ &= \frac{21}{\sqrt{21}} \\ &= \sqrt{21} \end{aligned}$$

So the distance between ℓ and Π is

$$\sqrt{21}$$

(iii) Use the reflection formulae for a plane $\Pi : \mathbf{r} \cdot \mathbf{n} = d$.

For a point,

$$\mathbf{p}' = \mathbf{p} - 2 \frac{\mathbf{p} \cdot \mathbf{n} - d}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

For a direction vector,

$$\mathbf{b}' = \mathbf{b} - 2 \frac{\mathbf{b} \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

Here

$$\mathbf{n} = \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix}, \quad d = -9$$

First reflect the direction vector of ℓ :

$$\mathbf{b} \cdot \mathbf{n} = 0$$

so

$$\begin{aligned}\mathbf{b}' &= \mathbf{b} - 2 \frac{0}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n} \\ &= \mathbf{b} \\ &= \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}\end{aligned}$$

So the direction is unchanged.

Now reflect the point $P = (4, 2, -3)$ on ℓ :

$$\begin{aligned}\mathbf{n} \cdot \mathbf{n} &= 1^2 + (-2)^2 + (-4)^2 = 21 \\ \mathbf{p} \cdot \mathbf{n} - d &= 12 - (-9) = 21\end{aligned}$$

Hence

$$\begin{aligned}\mathbf{p}' &= \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} - 2 \left(\frac{21}{21} \right) \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ -8 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 6 \\ 5 \end{pmatrix}\end{aligned}$$

Therefore the reflected line is

$$\mathbf{r} = \begin{pmatrix} 2 \\ 6 \\ 5 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

So an equation of the reflected line is

$$\mathbf{r} = \begin{pmatrix} 2 \\ 6 \\ 5 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

5. The line ℓ_1 has vector equation

$$\mathbf{r} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

The plane Π_1 passes through the point $(2, 0, 0)$ and has normal vector

$$\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

(a) Find the point of intersection of ℓ_1 and Π_1 .

[2]

The line ℓ_2 is the reflection of the line ℓ_1 in the plane Π_1 .

(b) Show that a vector equation for the line ℓ_2 is

$$\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

where μ is a scalar parameter.

[5]

The plane Π_2 contains the line ℓ_1 and the line ℓ_2 .

(c) Determine a vector equation for the line of intersection of Π_1 and Π_2 .

[2]

Solution

(a) The plane Π_1 passes through $(2, 0, 0)$ and has normal vector $\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$, so its equation is

$$\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \right) = 0$$

Hence

$$3(x - 2) - y + 2z = 0$$

so

$$3x - y + 2z = 6$$

A general point on ℓ_1 is

$$(x, y, z) = (1 + 2\lambda, 4 + \lambda, \lambda)$$

Substituting into the plane equation:

$$3(1 + 2\lambda) - (4 + \lambda) + 2\lambda = 6$$

$$3 + 6\lambda - 4 - \lambda + 2\lambda = 6$$

$$7\lambda - 1 = 6$$

$$7\lambda = 7$$

$$\lambda = 1$$

So the point of intersection is

$$(1 + 2, 4 + 1, 1) = (3, 5, 1)$$

Therefore ℓ_1 meets Π_1 at $(3, 5, 1)$.

(b) Let

$$\mathbf{a} = \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{n} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

so that

$$\ell_1 : \mathbf{r} = \mathbf{a} + \lambda \mathbf{u}$$

Since Π_1 passes through $(2, 0, 0)$,

$$\Pi_1 : \mathbf{r} \cdot \mathbf{n} = 6$$

because

$$\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 6$$

Using the reflection formulae for a plane $\mathbf{r} \cdot \mathbf{n} = 6$,

$$\mathbf{p}' = \mathbf{p} - 2 \frac{\mathbf{p} \cdot \mathbf{n} - 6}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

for a point, and

$$\mathbf{u}' = \mathbf{u} - 2 \frac{\mathbf{u} \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

for a direction vector.

First reflect the point $\mathbf{a}$.

$$\begin{aligned} \mathbf{a} \cdot \mathbf{n} &= \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \\ &= 3 - 4 + 0 \\ &= -1 \end{aligned}$$

Also,

$$\mathbf{n} \cdot \mathbf{n} = 3^2 + (-1)^2 + 2^2 = 14$$

So

$$\begin{aligned} \mathbf{a}' &= \mathbf{a} - 2 \frac{\mathbf{a} \cdot \mathbf{n} - 6}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n} \\ &= \mathbf{a} - 2 \frac{-1 - 6}{14} \mathbf{n} \\ &= \mathbf{a} - 2 \left(\frac{-7}{14} \right) \mathbf{n} \\ &= \mathbf{a} + \mathbf{n} \\ &= \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} \end{aligned}$$

Now reflect the direction vector $\mathbf{u}$.

$$\begin{aligned} \mathbf{u} \cdot \mathbf{n} &= \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \\ &= 6 - 1 + 2 \\ &= 7 \end{aligned}$$

Hence

$$\begin{aligned}\mathbf{u}' &= \mathbf{u} - 2 \frac{\mathbf{u} \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n} \\ &= \mathbf{u} - 2 \frac{7}{14} \mathbf{n} \\ &= \mathbf{u} - \mathbf{n} \\ &= \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}\end{aligned}$$

Therefore the reflected line ℓ_2 has vector equation

$$\mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

which is the required result.

(c) From part (a), the point

$$P = (3, 5, 1)$$

lies on Π_1 and on ℓ_1 .

Since $\ell_1 \subset \Pi_2$, the point P also lies on Π_2 . So the line of intersection of Π_1 and Π_2 passes through P .

A direction vector of ℓ_1 is

$$\mathbf{u}_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

and a direction vector of ℓ_2 is

$$\mathbf{u}_2 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

Because Π_2 contains both lines, it contains both direction vectors, so it also contains

$$\mathbf{u}_1 + \mathbf{u}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$$

To check that this is also parallel to Π_1 , take the dot product with the normal to Π_1 :

$$\begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 3 - 3 + 0 = 0$$

So $\begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$ is a direction vector for the line $\Pi_1 \cap \Pi_2$.

Hence a vector equation for the line of intersection is

$$\mathbf{r} = \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}$$

where t is a scalar parameter.

6. The triangle ABC lies in the plane Π . The vertices are

$$A(1, 0, 0), B(2, 2, 1) \text{ and } C(1, 1, c)$$

where c is a constant.

(a) Find, in terms of c , $\overrightarrow{BA} \times \overrightarrow{BC}$. [3]

Given that $\overrightarrow{BA} \times \overrightarrow{BC} = d\mathbf{i} + 2\mathbf{j} - \mathbf{k}$, where d is a constant,

(b) find the value of c and show that $d = -3$. [2]

(c) find an equation of Π in the form $\mathbf{r} \cdot \mathbf{n} = p$, where p is a constant. [3]

The point S has position vector $3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$. The point S' is the image of S under reflection in Π .

(d) Find the position vector of S' . [5]

Solution

(a) First find the two side vectors from B :

$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = (1, 0, 0) - (2, 2, 1) = (-1, -2, -1)$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = (1, 1, c) - (2, 2, 1) = (-1, -1, c-1)$$

Now take the cross product:

$$\begin{aligned} \overrightarrow{BA} \times \overrightarrow{BC} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -2 & -1 \\ -1 & -1 & c-1 \end{vmatrix} \\ &= \mathbf{i}((-2)(c-1) - (-1)(-1)) - \mathbf{j}((-1)(c-1) - (-1)(-1)) + \mathbf{k}((-1)(-1) - (-2)(-1)) \\ &= \mathbf{i}(-2c+2-1) - \mathbf{j}(-c+1-1) + \mathbf{k}(1-2) \\ &= (1-2c)\mathbf{i} + c\mathbf{j} - \mathbf{k} \end{aligned}$$

So,

$$\overrightarrow{BA} \times \overrightarrow{BC} = (1-2c)\mathbf{i} + c\mathbf{j} - \mathbf{k}$$

(b) We are given that

$$\overrightarrow{BA} \times \overrightarrow{BC} = d\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

From part (a),

$$\overrightarrow{BA} \times \overrightarrow{BC} = (1-2c)\mathbf{i} + c\mathbf{j} - \mathbf{k}$$

Comparing coefficients of $\mathbf{j}$,

$$c = 2$$

Then comparing coefficients of $\mathbf{i}$,

$$d = 1 - 2c = 1 - 2(2) = 1 - 4 = -3$$

Hence,

$$c = 2 \quad \text{and} \quad d = -3$$

(c) A normal vector to the plane Π is

$$\mathbf{n} = \overrightarrow{BA} \times \overrightarrow{BC}$$

Using $c = 2$, part (a) gives

$$\mathbf{n} = (1 - 2(2), 2, -1) = (-3, 2, -1)$$

So we can take

$$\mathbf{n} = -3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$$

The plane has equation $\mathbf{r} \cdot \mathbf{n} = p$. Since point $A(1, 0, 0)$ lies on Π ,

$$\begin{aligned} p &= \overrightarrow{OA} \cdot \mathbf{n} \\ &= (1, 0, 0) \cdot (-3, 2, -1) \\ &= -3 \end{aligned}$$

Therefore the equation of Π is

$$\mathbf{r} \cdot (-3\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = -3$$

Equivalently, in Cartesian form,

$$-3x + 2y - z = -3$$

(d) Let the position vector of S be

$$\mathbf{p} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k} = (3, 1, 2)$$

From part (c), the plane is

$$\mathbf{r} \cdot \mathbf{n} = d$$

with

$$\mathbf{n} = (-3, 2, -1), \quad d = -3$$

Using the reflection formula for a point in a plane,

$$\mathbf{p}' = \mathbf{p} - 2 \frac{\mathbf{p} \cdot \mathbf{n} - d}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

We now calculate the required quantities:

$$\begin{aligned} \mathbf{p} \cdot \mathbf{n} &= (3, 1, 2) \cdot (-3, 2, -1) \\ &= 3(-3) + 1(2) + 2(-1) \\ &= -9 + 2 - 2 \\ &= -9 \end{aligned}$$

$$\begin{aligned} \mathbf{n} \cdot \mathbf{n} &= (-3)^2 + 2^2 + (-1)^2 \\ &= 9 + 4 + 1 \\ &= 14 \end{aligned}$$

So

$$\begin{aligned} \mathbf{p}' &= \mathbf{p} - 2 \frac{-9 - (-3)}{14} \mathbf{n} \\ &= \mathbf{p} - 2 \frac{-6}{14} \mathbf{n} \\ &= \mathbf{p} + \frac{12}{14} \mathbf{n} \\ &= \mathbf{p} + \frac{6}{7} \mathbf{n} \end{aligned}$$

Substituting $\mathbf{p} = (3, 1, 2)$ and $\mathbf{n} = (-3, 2, -1)$,

$$\begin{aligned}\mathbf{p}' &= (3, 1, 2) + \frac{6}{7}(-3, 2, -1) \\ &= \left(3 - \frac{18}{7}, 1 + \frac{12}{7}, 2 - \frac{6}{7}\right) \\ &= \left(\frac{21}{7} - \frac{18}{7}, \frac{7}{7} + \frac{12}{7}, \frac{14}{7} - \frac{6}{7}\right) \\ &= \left(\frac{3}{7}, \frac{19}{7}, \frac{8}{7}\right)\end{aligned}$$

Hence the position vector of S' is

$$\frac{3}{7}\mathbf{i} + \frac{19}{7}\mathbf{j} + \frac{8}{7}\mathbf{k}$$

7. The line ℓ has equation

$$\mathbf{r} = (\mathbf{i} + \mathbf{k}) + \lambda(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$$

where λ is a scalar parameter, and the plane Π has equation

$$\mathbf{r} \cdot (2\mathbf{i} - \mathbf{j} + \mathbf{k}) = 9$$

(a) Find the coordinates of the point of intersection of ℓ and Π . [4]

The perpendicular to Π from the point $A(0, 4, 1)$ meets Π at the point B .

(b) Verify that the coordinates of B are $(4, 2, 3)$. [3]

The point $A(0, 4, 1)$ is reflected in the plane Π to give the image point A' .

(c) Find the coordinates of the point A' . [2]

(d) Find an equation for the line obtained by reflecting the line ℓ in the plane Π , giving your answer in the form

$$\mathbf{r} \times \mathbf{a} = \mathbf{b} \quad [4]$$

Solution

(a) Write the line ℓ in component form:

$$\mathbf{r} = (1, 0, 1) + \lambda(1, 2, 3)$$

so

$$(x, y, z) = (1 + \lambda, 2\lambda, 1 + 3\lambda)$$

The plane Π has equation

$$\mathbf{r} \cdot (2, -1, 1) = 9$$

so in Cartesian form,

$$2x - y + z = 9$$

Substituting the line coordinates into the plane equation:

$$2(1 + \lambda) - 2\lambda + (1 + 3\lambda) = 9$$

$$2 + 2\lambda - 2\lambda + 1 + 3\lambda = 9$$

$$3 + 3\lambda = 9$$

$$3\lambda = 6$$

$$\lambda = 2$$

Hence the point of intersection is

$$(x, y, z) = (1 + 2, 2 \cdot 2, 1 + 3 \cdot 2) = (3, 4, 7)$$

Therefore, the line ℓ meets the plane Π at $(3, 4, 7)$.

(b) A normal vector to the plane Π is

$$\mathbf{n} = (2, -1, 1)$$

So the perpendicular through $A(0, 4, 1)$ has equation

$$\mathbf{r} = (0, 4, 1) + t(2, -1, 1)$$

Hence points on this line have coordinates

$$(x, y, z) = (2t, 4 - t, 1 + t)$$

Substituting into the plane equation $2x - y + z = 9$:

$$2(2t) - (4 - t) + (1 + t) = 9$$

$$4t - 4 + t + 1 + t = 9$$

$$6t - 3 = 9$$

$$6t = 12$$

$$t = 2$$

So

$$B = (2 \cdot 2, 4 - 2, 1 + 2) = (4, 2, 3)$$

Hence the coordinates of B are verified to be $(4, 2, 3)$.

(c) Using the reflection formula for a point in the plane $\Pi : \mathbf{r} \cdot \mathbf{n} = d$,

$$\mathbf{p}' = \mathbf{p} - 2 \frac{\mathbf{p} \cdot \mathbf{n} - d}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

Here

$$\mathbf{p} = (0, 4, 1), \quad \mathbf{n} = (2, -1, 1), \quad d = 9$$

First calculate:

$$\mathbf{p} \cdot \mathbf{n} = 0 \cdot 2 + 4(-1) + 1 \cdot 1 = -3, \quad \mathbf{n} \cdot \mathbf{n} = 2^2 + (-1)^2 + 1^2 = 6$$

Therefore

$$\begin{aligned} \mathbf{p}' &= (0, 4, 1) - 2 \frac{-3 - 9}{6} (2, -1, 1) \\ &= (0, 4, 1) - 2 \left(\frac{-12}{6} \right) (2, -1, 1) \\ &= (0, 4, 1) + 4(2, -1, 1) \\ &= (8, 0, 5) \end{aligned}$$

So the image point is $A' = (8, 0, 5)$.

(d) Let P be the point where ℓ meets Π . From part (a),

$$P = (3, 4, 7)$$

Since P lies in the mirror plane, it stays fixed under reflection.

The direction vector of ℓ is

$$\mathbf{d} = (1, 2, 3)$$

Using the reflection formula for a direction vector,

$$\mathbf{d}' = \mathbf{d} - 2 \frac{\mathbf{d} \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}} \mathbf{n}$$

with $\mathbf{n} = (2, -1, 1)$.

First calculate:

$$\mathbf{d} \cdot \mathbf{n} = 1 \cdot 2 + 2(-1) + 3 \cdot 1 = 3, \quad \mathbf{n} \cdot \mathbf{n} = 6$$

So

$$\begin{aligned}\mathbf{d}' &= (1, 2, 3) - 2 \left(\frac{3}{6} \right) (2, -1, 1) \\ &= (1, 2, 3) - (2, -1, 1) \\ &= (-1, 3, 2)\end{aligned}$$

Hence the reflected line ℓ' is

$$\mathbf{r} = (3, 4, 7) + \mu(-1, 3, 2)$$

To write this in the form $\mathbf{r} \times \mathbf{a} = \mathbf{b}$, take

$$\mathbf{a} = (-1, 3, 2)$$

Then

$$\mathbf{r} = (3, 4, 7) + \mu\mathbf{a}$$

implies

$$\mathbf{r} \times \mathbf{a} = ((3, 4, 7) + \mu\mathbf{a}) \times \mathbf{a} = (3, 4, 7) \times \mathbf{a}$$

since $\mathbf{a} \times \mathbf{a} = \mathbf{0}$.

Now calculate

$$\begin{aligned}(3, 4, 7) \times (-1, 3, 2) &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 4 & 7 \\ -1 & 3 & 2 \end{vmatrix} \\ &= \mathbf{i}(4 \cdot 2 - 7 \cdot 3) - \mathbf{j}(3 \cdot 2 - 7(-1)) + \mathbf{k}(3 \cdot 3 - 4(-1)) \\ &= \mathbf{i}(8 - 21) - \mathbf{j}(6 + 7) + \mathbf{k}(9 + 4) \\ &= -13\mathbf{i} - 13\mathbf{j} + 13\mathbf{k}\end{aligned}$$

Therefore an equation of the reflected line is

$$\mathbf{r} \times (-\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}) = -13\mathbf{i} - 13\mathbf{j} + 13\mathbf{k}$$

So

$$\mathbf{a} = (-1, 3, 2), \quad \mathbf{b} = (-13, -13, 13)$$